

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0170

Scoggin 138 kV
January 20, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0170. The Interconnection Customer proposes a 186.0 MW battery energy storage facility interconnecting at the Scoggin 138 kV substation in the Pennsylvania area. The request was received on October 31, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 127 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 26 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$34,526,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Carbon Ctr 500-345 kV Auto
Contingency: Blackrock-Shawville Sta 345kV
Loading: 141.6% (Rating: 523 MVA)
DFAX: 5.9%
Pre-project loading: 139.5%
2. Facility: Piney Cree Project 500/345 kV Auto
Contingency: Piney Crk Project-Hilltop Substation 500kV
Loading: 140.8% (Rating: 776 MVA)
DFAX: 6.5%
Pre-project loading: 139.3%
3. Facility: Silver King-Betzwood 138kV
Contingency: Scoggin Substation 345/138kV Transformer Bank
Loading: 138.0% (Rating: 240 MVA)
DFAX: 41.0%
Pre-project loading: 106.2%
4. Facility: Silver King-Quechan 138kV
Contingency: Scoggin 345 kV/138 kV TX
Loading: 127.0% (Rating: 153 MVA)
DFAX: 45.6%
Pre-project loading: 71.7%
5. Facility: Koenig Lane-Scoggin 138kV
Contingency: Silver King-Betzwood 138kV L
Loading: 125.2% (Rating: 262 MVA)
DFAX: 42.3%
Pre-project loading: 95.2%

6. Facility: Clear Springs 500/345 kV Transformer Bank
Contingency: Burns-Hills Chapel 345kV L
Loading: 121.0% (Rating: 883 MVA)
DFAX: 6.6%
Pre-project loading: 119.6%
7. Facility: Hilltop 500/345kV Transformer
Contingency: Hilltop-Taleneergy Martins Creek Llc Williamsport 500kV
Loading: 120.3% (Rating: 1,001 MVA)
DFAX: 7.4%
Pre-project loading: 118.9%
8. Facility: Curecanti-Moser Generating Sta 345kV
Contingency: Bixby Hill-Tombstone 500kV
Loading: 118.1% (Rating: 1,413 MVA)
DFAX: 18.9%
Pre-project loading: 115.6%
9. Facility: Talenenergy Martins Crk Llc Williamsport 500/345 kV Transformer
Contingency: Hilltop Substation 500/345kV Auto
Loading: 116.2% (Rating: 1,035 MVA)
DFAX: 7.3%
Pre-project loading: 114.9%
10. Facility: Talenenergy Martins Creek Llc Harwood Substation 500/1 kV XFMR Bank 1
Contingency: Feida 500/345kV XFMR
Loading: 116.2% (Rating: 1,103 MVA)
DFAX: 5.9%
Pre-project loading: 115.2%
11. Facility: Eliza Crk-Ontelaunee Energy Ctr 345kV Line
Contingency: Tombstone Substation-Bixby Hill 500kV
Loading: 115.9% (Rating: 1,213 MVA)
DFAX: 17.2%
Pre-project loading: 113.2%
12. Facility: Little Chute-Talenenergy Martins Crk Llc Harrisburg 345kV line
Contingency: Fort Humbug-Coronet 500kV
Loading: 114.9% (Rating: 1,230 MVA)
DFAX: 7.6%
Pre-project loading: 113.8%
13. Facility: W Campus-Fort Humbug 345kV L
Contingency: Orchard Rd-Ft Humbug 500kV
Loading: 110.8% (Rating: 1,720 MVA)
DFAX: 9.2%
Pre-project loading: 109.8%
14. Facility: Pueblo Xtra Substation 500 kV/230 kV Transformer
Contingency: Pueblo Xtra-Boca Raton 500kV
Loading: 109.5% (Rating: 766 MVA)
DFAX: 5.2%
Pre-project loading: 108.3%
15. Facility: Felda-W Gallatin 345kV
Contingency: Wallenpaupack Substation 500/345 kV Auto
Loading: 109.2% (Rating: 1,237 MVA)
DFAX: 5.8%
Pre-project loading: 108.3%
16. Facility: W Gallatin-Walnut 345kV
Contingency: Wallenpaupack 500/345kV Transformer Bank

Loading: 108.8% (Rating: 1,180 MVA)

DFAX: 5.9%

Pre-project loading: 107.8%

17. Facility: Curecanti-Eliza Creek 345kV

Contingency: Tombstone-Bixby Hill 500kV

Loading: 107.5% (Rating: 1,326 MVA)

DFAX: 19.3%

Pre-project loading: 104.8%

18. Facility: Betzwood-Neset 138kV

Contingency: Scoggin 345 kV/138 kV TX

Loading: 107.1% (Rating: 236 MVA)

DFAX: 26.1%

Pre-project loading: 86.5%

19. Facility: Talenenergy Martins Crk Llc Harwood-Felida 345kV line

Contingency: Felida 500/345kV Auto

Loading: 106.4% (Rating: 1,209 MVA)

DFAX: 5.4%

Pre-project loading: 105.5%

20. Facility: Hills Chapel-Burns 345kV line

Contingency: Clear Springs 500/345kV TX

Loading: 105.8% (Rating: 1,207 MVA)

DFAX: 5.4%

Pre-project loading: 104.9%

21. Facility: Talenenergy Martins Crk Llc Harwood 345-1 kV Auto

Contingency: Felida 500/345kV Autotransformer

Loading: 105.7% (Rating: 1,213 MVA)

DFAX: 5.9%

Pre-project loading: 104.8%

22. Facility: Sunrise Center-Wahiawa 345kV

Contingency: Sunrise Center-Merret 500kV L

Loading: 105.5% (Rating: 1,205 MVA)

DFAX: 5.1%

Pre-project loading: 104.7%

23. Facility: Frt Humbug 500/345kV Transformer Bank

Contingency: Orchard Rd-Ft Humbug 500kV

Loading: 105.5% (Rating: 925 MVA)

DFAX: 5.8%

Pre-project loading: 104.3%

24. Facility: Fort Humbug-Little Chute 345kV

Contingency: Ft Humbug-Coronet 500kV

Loading: 104.7% (Rating: 1,196 MVA)

DFAX: 5.7%

Pre-project loading: 103.9%

25. Facility: Wallenpaupack 500 kV/345 kV Transformer

Contingency: West Gallatin-Walnut 345kV

Loading: 103.3% (Rating: 1,351 MVA)

DFAX: 5.5%

Pre-project loading: 102.6%

26. Facility: Orchard Rd-Fort Humbug 500kV

Contingency: Ft Humbug-West Campus 345kV

Loading: 102.7% (Rating: 2,551 MVA)

DFAX: 13.4%

Pre-project loading: 101.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Carbon Ctr 500-345 kV Auto	Replace transformer with higher-rated unit	\$25,686,000	4% (\$1,088,000)
Piney Cree Project 500/345 kV Auto	Replace transformer with higher-rated unit	\$26,818,000	3% (\$846,000)
Silver King-Betzwood 138kV	Rebuild line with higher-capacity conductor	\$3,665,000	100% (\$3,665,000)
Silver King-Quechan 138kV	Rebuild line with higher-capacity conductor	\$3,379,000	100% (\$3,379,000)
Koenig Lane-Scoggin 138kV	Rebuild line with higher-capacity conductor	\$1,936,000	100% (\$1,936,000)
Clear Springs 500 kV/345 kV Transformer	Replace transformer with higher-rated unit	\$29,163,000	4% (\$1,228,000)
Hilltop 500-345 kV TX	Replace transformer with higher-rated unit	\$33,657,000	4% (\$1,437,000)
Moser Generating Sta-Curecanti 345kV	Rebuild line with higher-capacity conductor	\$26,399,000	7% (\$1,953,000)
Talenenergy Martins Creek Llc Williamsport 500-345 kV Autotransformer	Replace transformer with higher-rated unit	\$33,497,000	5% (\$1,525,000)
Talenenergy Martins Creek Llc Harwood 500/1kV XFMR	Replace transformer with higher-rated unit	\$25,247,000	4% (\$1,079,000)
Ontelaunee Enery Center-Eliza Crk 345kV	Rebuild line with higher-capacity conductor	\$16,743,000	8% (\$1,346,000)
Talenenergy Martins Creek Llc Harrisburg-Little Chute 345kV	Rebuild line with higher-capacity conductor	\$20,935,000	4% (\$851,000)
W Campus-Ft Humbug 345kV line	Rebuild line with higher-capacity conductor	\$68,963,000	4% (\$2,669,000)
Pueblo Xtra 500/230kV Auto	Replace transformer with higher-rated unit	\$41,779,000	3% (\$1,067,000)
Felda-W Gallatin 345kV	Reconductor line	\$10,570,000	4% (\$437,000)
W Gallatin-Walnut 345kV	Reconductor line	\$13,648,000	4% (\$524,000)
Eliza Creek-Curecanti 345kV Line	Reconductor line	\$7,674,000	8% (\$587,000)
Betzwood-Neset 138kV	Reconductor line	\$1,155,000	100% (\$1,155,000)
Talenenergy Martins Crk Llc Harwood-Felida 345kV	Reconductor line	\$5,738,000	4% (\$247,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hills Chapel-Burns 345kV	Reconductor line	\$21,900,000	4% (\$865,000)
Talenenergy Martins Crk Llc Harwood Substation 345-1 kV Autotransformer	Replace transformer with higher-rated unit	\$14,631,000	4% (\$625,000)
Sunrise Center Substation-Wahiawa 345kV	Reconductor line	\$8,155,000	4% (\$370,000)
Frt Humbug 500/345kV Transformer Bank	Replace transformer with higher-rated unit	\$40,362,000	5% (\$1,890,000)
Little Chute-Ft Humbug 345kV	Reconductor line	\$42,142,000	4% (\$1,636,000)
Wallenpaupack 500 kV/345 kV TX	Replace transformer with higher-rated unit	\$32,928,000	4% (\$1,302,000)
Ft Humbug-Orchard Road 500kV	Reconductor line	\$23,380,000	4% (\$819,000)
	Total allocated to project		\$34,526,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.